

Online Appendix

Sourcing under Sanctions: Judicial Urgency and Pharmaceutical Procurement Costs

Darcio Genicolo-Martins^a, Paulo Furquim de Azevedo^a

^a*Inspere Institute of Education and Research, São Paulo, Brazil*

Supplementary material for the short paper. All empirical values are generated by the v9 build pipeline and imported through `values.tex`, generated tables, or generated figures.

A. Data Construction and Classifier Validation

The procurement data come from BEC pharmaceutical purchases in group 65 during 2009–2019. The construction starts from 764,362 classified purchase orders and links procurement fields to litigation and administrative-process markers. Ordinary purchases are those without urgent judicial or administrative markers. Administrative urgent purchases are requests processed through the health agency’s administrative channel. Litigated urgent purchases are purchases tied to court-mandated delivery.

The classifier combines four signals: machine-learning predictions, structured JSON flags, regex patterns, and position-based checks in the tender notice. The production rule uses weights 3, 3, 1, and 1 for these signals and classifies a regime when the vote reaches 2. Boilerplate language is filtered using 10 exclusion patterns.

Table A.1: Purchase-regime classifier validation.

Validation object	Precision / agreement	Recall	F1
Exact three-class agreement	98.6%	–	–
Judicial class, ground truth	0.87	1.00	0.93
Administrative class, ground truth	0.92	0.99	0.96
Urgent-class macro-F1	–	–	0.94
Judicial class, ML hold-out	–	–	0.93
Administrative class, ML hold-out	–	–	0.94

Notes: Ground-truth validation uses 179,148 labeled purchase orders; exact agreement covers 176,713 matches. The ML hold-out test set contains 34,191 observations. Precision and recall entries are rounded by the generator from the production classification report where reported.

Validation against 179,148 ground-truth observations gives 98.6% exact agreement. The urgent-class F1 scores are 0.93 for judicial purchases and 0.96 for administrative purchases; the macro-F1 over these urgent classes is 0.94. The held-out machine-learning test set contains 34,191 observations and produces F1 scores of 0.93 and 0.94. Because the empirical comparisons use classified regimes, remaining false positives and false negatives would attenuate regime contrasts if they are not systematically correlated with residual prices within controls.

B. Sample Construction and Balance

The main-paper sample table reconciles the core samples. The full BEC file contains 479,330 purchase-offer-item observations. The analysis sample contains 226,305 observations, and the main winners-only price sample contains 196,883 observations. The urgent panel contains 56,803 winner observations used for the under-the-gun contrast. The firm-buyer-item triple sample contains 4,573 observations in 1,206 triples.

Table B.1: Sample means by purchase type.

	Ordinary	Administrative	Litigated
Observations (winners)	343,393	16,985	46,212
Log reference price	2.109	0.960	1.578
Log negotiated price	1.596	0.543	1.050
Quantity (units)	56,280	794,534	61,903
Number of bidding firms	3.159	2.394	2.191
Tender success rate	84.1%	85.7%	90.7%

Means computed on winners (POIs with an accepted winning bid). Reference and negotiated prices are in logs; quantity in units; tender success rate is the share of POIs that close with an accepted winning bid. Sample period: 2009–2019. Source: BEC-SP, SES/SP S-CODES.

Administrative and litigated urgent purchases are not balanced by design. Administrative purchases are larger and differ in reference prices and formulary status. This imbalance is why the main design combines selection bounds, within-firm pricing tests, and direct sourcing evidence rather than treating the administrative channel as randomly assigned.

C. Selection Bounds

The administrative channel screens requests before purchase. Let $A = 1$ denote administrative urgent procurement and $A = 0$ denote litigated urgent procurement. The main under-the-gun

Table B.2: Within-Item Balance: Administrative vs. Litigated

Covariate	Mean (Admin)	Mean (Lit)	Raw diff. (A-L)	Within-item diff.	
				Coef	(SE)
SUS basic (MEDICAMENTO flag)	0.718	0.892	-0.174	0.000 ^{NA}	(NaN)
Electronic auction (pregão)	0.984	0.941	0.043	0.041 ^{**}	(0.017)
Log quantity (bulk size signal)	7.642	6.180	1.462	0.866 [*]	(0.502)
Log reference price	0.835	1.401	-0.566	-0.339 ^{***}	(0.093)
Successful tender	0.869	0.914	-0.045	-0.024	(0.022)
Late period (year ≥ 2014)	0.565	0.623	-0.058	-0.045	(0.041)
Large PBU (above-median)	0.895	0.810	0.085	0.083	(0.073)
Observations	63,426				
Items	2,370				

Notes: Within-item balance between administrative ($Admin = 1$) and litigated ($Admin = 0$) urgent purchases, restricted to items with both types present. Raw difference is the unconditional mean gap. Within-item coefficient is β from $x_{i,g,t} = \beta Admin_{i,g,t} + \gamma_g + \varepsilon_{i,g,t}$, where γ_g is an item fixed effect and standard errors are clustered at the PBU level. A coefficient close to zero means that, within a given item, the administrative and litigated draws are balanced on the covariate. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table B.3: Representativeness of both-types cells: cells with both administrative and litigated urgent purchases (5,581 cells, 31.8%) vs. singleton cells (11,972).

Variable	Both-types cells	Singleton cells	p-value
Log reference price	1.233	1.761	0.000
Log negotiated price	0.793	1.326	0.000
Log quantity	6.560	6.324	0.000
N. bidding firms	2.181	2.401	0.000
SUS formulary share	0.867	0.725	0.000
Revers-auction (pregão) share	0.955	0.952	0.039

Means computed at the observation level within each cell type. p-values from two-sample t-tests. Both-types cells provide the within-cell variation that identifies the itemtime-year-month UTG specification. Differences in modality, supplier-base depth, and SUS formulary status indicate that the within-cell estimator is identified off a non-random subsample of urgent items.

coefficient is the administrative-minus-litigated log price coefficient. A negative coefficient means litigated purchases are more expensive.

The Manski-Lee procedure trims the overrepresented administrative group within item-by-year-by-buyer strata. If administrative cases are selected among requests that would otherwise appear under litigation, monotone selection implies that trimming one tail of the administrative price distribution bounds the counterfactual litigated comparison. Trimming the top tail produces one bound; trimming the bottom tail produces the other. The preferred bounds imply a litigated-over-administrative price gap of [15.9%, 21.1%].

The mean trimming rate is 26.9%, with a maximum of 100.0%. Administrative-excess trimming occurs in 12,038 of 44,654 strata. The sign convention in all tables is administrative minus litigated

Table C.1: Manski-Lee bounds on the under-the-gun coefficient.

Specification	log negotiated price		implied		N
	Coef. on Admin	SE	Lit-vs-Admin (%)		
Naive UTG (item + year + PBU FE)	-0.259	0.092	29.530		61620
Lee lower bound (admin top-trim)	-0.192	0.095	15.924		45624
Lee upper bound (admin bottom-trim)	-0.148	0.097	21.123		45624

Trimming applied within itemtime-yearmonthPBU strata where the admin sample exceeds the litigated count, by the differential proportion. Lower bound trims the top of the admin price distribution; upper bound trims the bottom. Standard errors clustered at PBU level.

Table C.2: Heckman selection-corrected UTG estimate.

Term	Estimate	SE
First stage: pre-period log ref price	-0.051	0.007
First stage: pre-period mean age	0.000	0.000
First stage: pre-period mean firm	0.005	0.020
First stage: pre-period α orders (mcf)	0.000	0.000
First stage: SUS formulary	-0.265	0.042
Second stage: Admin conf	0.817	4.278
Second stage: inverse Mills ratio	-0.053	2.037
Second stage: implied lit-cost-admin (%)	-55.841	NA
Heckit ML ρ	1.000	NA

First stage: profile of admin on pre-period from characteristics. pre_α orders is excluded from the outcome equation as the exclusion restriction. Second stage: fits with firm, year, and PBU FE. Mills ratio constructed from the first-stage prediction. Standard errors clustered at the PBU level. Heckit ML rho reported as a sanity cross-check; identification of rho without FE is fragile.

for coefficients and litigated over administrative for percentage gaps.

The Heckman-style correction is reported only as a diagnostic. The first-stage exclusions are weak for this institutional comparison, and the second-stage estimate is not informative. The main text therefore relies on trimming bounds and mechanism tests.

D. Additional Robustness

This appendix section collects robustness checks that support the main mechanism without extending the short paper. Alternative fixed effects, firm fixed effects, placebo checks, and heterogeneity by market depth all point to the same interpretation: the cost margin is not a broad same-firm markup in deep urgent markets.

Table D.1: Supplier Fixed Effects: Demand vs Supply Side			
	bid_price_log		
	(1)	(2)	(3)
Urgent Purchase	0.0513*** (0.0153)	0.0245 (0.0159)	0.0109 (0.0179)
Log Quantity			-0.3931*** (0.0325)
Observations	196,883	196,642	196,642
R ²	0.86796	0.88083	0.92157
Within R ²	0.00022	5.41×10^{-5}	0.34189
item_id fixed effects	✓	✓	✓
year_n fixed effects	✓	✓	✓
pbu_id fixed effects	✓	✓	✓
firm_f fixed effects		✓	✓

DV: log negotiated price. Sample: winning bids for items with both ordinary and litigated purchases. Column (1): item + year + PBU FE. Column (2) adds firm FE. Column (3) adds log quantity. SE clustered at PBU level.

The within-firm heterogeneity is central. The baseline coefficient is 0.035 with standard error 0.041. The above-median quantity coefficient is -0.005; the below-median quantity coefficient is +0.066. The SUS-formulary coefficient is -0.001; the non-formulary coefficient is +0.101. The

Table D.2: Placebo Test: Items Never Subject to Litigation

	bid_price_log		bid_price_ref_log	
	(1)	(2)	(3)	(4)
Urgent Purchase	-0.0204 (0.0316)	0.0513*** (0.0153)	-0.0306 (0.0447)	0.0305** (0.0141)
Observations	39,283	196,883	46,874	226,278
R ²	0.83364	0.86796	0.81121	0.85183
Within R ²	7.03×10^{-6}	0.00022	1.73×10^{-5}	7.43×10^{-5}
item_id fixed effects	✓	✓	✓	✓
year_n fixed effects	✓	✓	✓	✓
pbu_id fixed effects	✓	✓	✓	✓

Placebo sample: items with zero litigated purchases across 2009-2019. Main sample: items with at least one litigated and one ordinary purchase. DV: log price. Item + Year + PBU FE. SE clustered at PBU level.

Table D.3: Wild cluster bootstrap inference on the UTG coefficient.

Specification	$\hat{\beta}$	Asy. SE	t	p_{boot}	95% CI (log-points)
Preferred (item + year + PBU)	-0.259	0.092	-2.80	0.008	$[-0.469, -0.039]$
Tightest (item×year-month + PBU)	-0.283	0.116	-2.44	0.039	$[-0.549, -0.001]$

Notes: Rademacher wild cluster bootstrap on the Admin coefficient with $B = 999$ replicates and restricted-null residuals; clustering at the PBU level. Negative coefficients mean litigated purchases are more expensive than administrative ones, in log-points. Source: `analysis/44_wild_bootstrap.R`.

earlier-period coefficient is +0.117, while the later-period coefficient is -0.038.

E. Dynamic Evidence

Dynamic estimates diagnose timing but do not carry the main identification strategy. The BJS event-study estimate at exposure is 0.052 with standard error 0.018; the later estimate is 0.147 with standard error 0.026. Honest-DiD sensitivity does not survive deviations at the observed maximum pre-period scale, recorded as no.

F. Procurement-Cost Calculation

The procurement-cost calculation applies the Lee midpoint 18.5% to an admissibility share of 50.0% and annual litigated spending of \$300 M. This yields \$27.8 M per year, with a Lee-bound range from \$23.9 M to \$31.7 M. This is a fiscal procurement-cost calculation, not a welfare estimate.

Table D.4: Within firm-buyer-item null: robustness across subsamples.

Subsample	$\hat{\beta}_{\text{Admin}}$	SE	N
All triples (baseline)	0.035	0.041	4,573
Above-median quantity	-0.005	0.044	2,188
Below-median quantity	0.066***	0.025	2,027
SUS-formulary items	-0.001	0.026	2,540
Non-formulary items	0.101	0.064	2,033
Earlier period	0.117***	0.037	2,553
Later period	-0.038	0.052	1,492

Notes: Each row reports the within firm-buyer-item triple Admin coefficient on log negotiated price, with FBI and year fixed effects, PBU-clustered SEs. Subsamples are constructed from the triple sample (1,206 triples). Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table F.1: UTG reconciliation: observed log-price gap decomposed into mechanical bulk-discount prediction, within firm-buyer-item offset, and supplier composition residual. 1,206 triples observed in both regimes.

Component	Log price (column 2)	Log price (column 3)
Observed UTG (column 1)	-0.058	-0.058
Mechanical CT bulk discount + UTG gap	-0.007	-0.007
Within firm-buyer-item offset	0.041	0.041
Supplier composition residual	0.016	0.016

Table F.2: Selection-bias-corrected welfare bound on the procurement cost of judicial enforcement.

Component	Welfare
Observed UTG (column 1)	-0.058
Mechanical CT bulk discount + UTG gap	-0.007
Within firm-buyer-item offset	0.041
Supplier composition residual	0.016

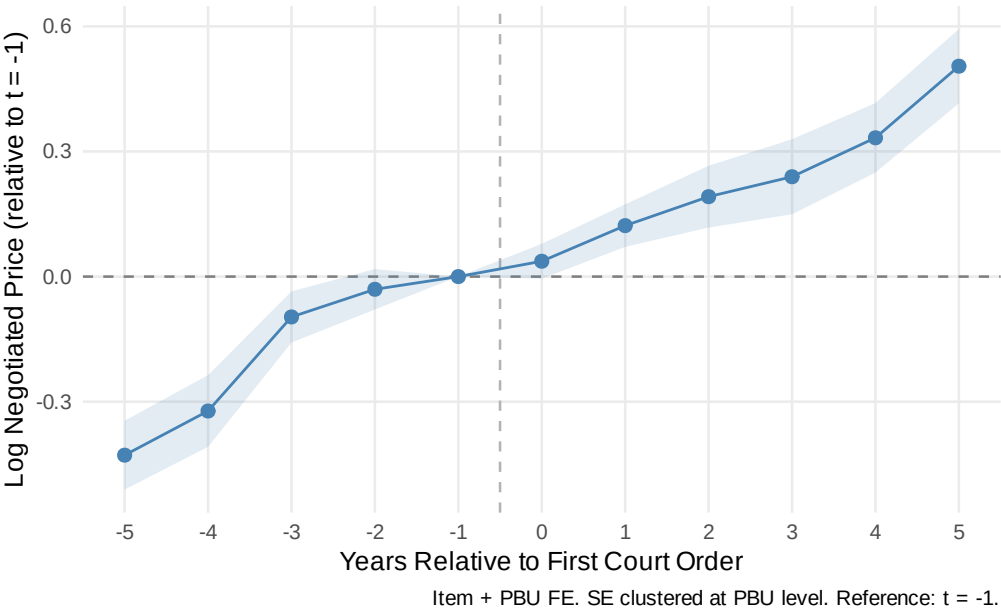


Figure E.1: BJS event-study diagnostic for item-level exposure. The figure is used to assess timing patterns; it is not the primary identification design.

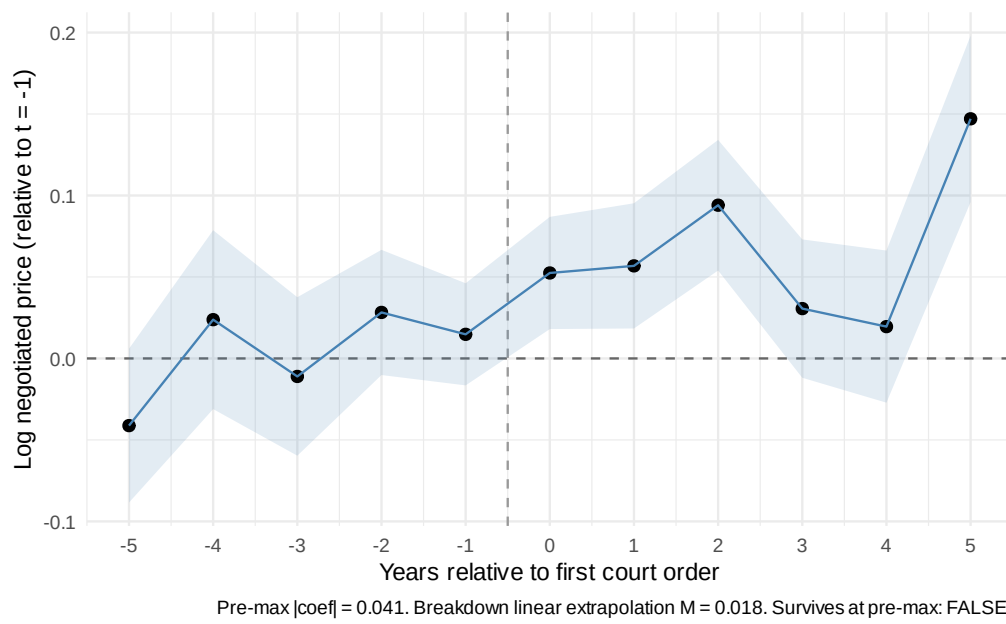


Figure E.2: Honest-DiD sensitivity diagnostic. The dynamic evidence is supportive, but the short paper's main claims rely on selection bounds, within-firm pricing tests, and sourcing evidence.